



Cambridge International AS & A Level

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FURTHER MATHEMATICS

9231/12

Paper 1 Further Pure Mathematics 1

October/November 2025

2 hours

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.



- 1 (a) Use standard results from the list of formulae (MF19) to find $\sum_{r=1}^n (r^3 - r)$ in terms of n , fully factorising your answer. [3]

[illegible]



- (b)** Express $\frac{r+3}{r^3-r}$ in the form $\frac{A}{r-1} + \frac{B}{r} + \frac{C}{r+1}$, where A , B and C are constants to be determined, and hence use the method of differences to find $\sum_{r=2}^n \frac{r+3}{r^3-r}$. [5]

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- (c) Deduce the value of $\sum_{r=2}^{\infty} \frac{r+3}{r^3-r}$. [1]





- 2** The cubic equation $x^3 + bx^2 + cx + d = 0$, where b , c and d are constants, has roots α , β and γ . It is given that

$$\begin{aligned}\alpha + \beta + \gamma &= 2, \\ \alpha^2 + \beta^2 + \gamma^2 &= 3, \\ \alpha^4 + \beta^4 + \gamma^4 &= 5.\end{aligned}$$

- (a)** Find the values of b and c .

[3]

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- 3 The sequence of positive numbers $u_1, u_2, u_3, \dots$ is such that $u_1 < 5$ and, for $n \geq 1$,

$$u_{n+1} = \frac{6u_n + 5}{u_n + 2}.$$

- (a) By considering $5 - u_{n+1}$, prove by mathematical induction that $u_n < 5$ for all positive integers n . [5]

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for letter height. The lines are evenly spaced across the entire page, providing a guide for consistent letter formation. There is no text or other markings on the page.



4 Let k and m be non-zero constants. The matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are given by

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \\ 1 & 1 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} k & 0 \\ 0 & m \end{pmatrix} \text{ and } \mathbf{C} = \begin{pmatrix} 2 & -1 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

(a) Give full details of the geometrical transformation in the x - y plane represented by the matrix $\mathbf{B}$ in each of the following cases.

(i) $m = 1$ [2]

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(ii) $m = k$ [2]

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(b) Show that the matrix $\mathbf{ABC}$ is singular. [6]

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[illegible]



5 The curve C has polar equation $r^2 = \tan 2\theta$, where $0 \leq \theta \leq \frac{1}{8}\pi$.

(a) Sketch C and state the greatest distance of a point on C from the pole.

[2]

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(b) Find the exact value of the area of the region bounded by C and the half-line $\theta = \frac{1}{8}\pi$.

[4]

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- (c) Show that C has Cartesian equation $x^4 - 2xy - y^4 = 0$ given that $0 \leq x \leq \cos\left(\frac{1}{8}\pi\right)$ and $0 \leq y \leq \sin\left(\frac{1}{8}\pi\right)$.

[4]

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- (d) Using your answer to (b), deduce the exact value of the area bounded by C , the x -axis and the line $x = \cos\left(\frac{1}{8}\pi\right)$.

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- 6 The plane Π has equation $x + 3y + 2z = 1$.

- (a) Find the perpendicular distance from the origin O to the plane Π . [2]

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Relative to O , the points A, B, C have position vectors

$$-\mathbf{j} + 2\mathbf{k}, \quad 2\mathbf{i} - \mathbf{k}, \quad 2\mathbf{i} - \mathbf{j} - \mathbf{k},$$

respectively.

- (b) Find the acute angle between the planes OAB and Π . [4]

[illegible]



(c) Find an equation for the common perpendicular to the lines OC and AB .

[8]

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7 The curve C has equation $y = \frac{x^2 + x + 1}{x + 1}$.

(a) Find the equations of the asymptotes of C . [3]

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(b) Find the coordinates of any stationary points on C . [3]

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(c) Sketch C .

[3]

(d) Sketch the curve with equation $y = \frac{|x|^2 + |x| + 1}{|x| + 1}$.

[2]





- (e) Find, in exact form, the set of values of x for which $\frac{|x|^2 + |x| + 1}{|x| + 1} < 3$. [3]

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Additional page

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